RV COLLEGE OF ENGINEERING
Bengaluru-560 059

REPORT ON

EXPERIENTIAL LEARNING / PROJECT BASED LEARNING

ACY 2023-24

THEME

QUANTUM MECHANICS

Title of the Project

Li-Fi (VLC)

Students Group

SL. No.	USN	Name	Prog.
1	1RV23AI039	Ishita Goyal	AIML
2	1RV23AS004	Akula Uday Kiran	ASE
3	1RV23AI015	Anamay Mittal	AIML
4	1RV23AS061	Tejas L	ASE

Project Evaluators

SL. No.	Name of the Evaluators (Enter the Names in Capitals)
1.	Dr TRIBIKRAM GUPTA
2.	Dr MOHANA
3.	Dr ABHAY A DESHPANDE
4.	Prof P L RAJASHEKAR


Li-Fi (VLC)

INTRODUCTION

Visible light communication (VLC) is a wireless method that enables high-speed transmission of data with visible light. This data is transmitted by modulating the intensity of light given off by a light source. The signal is received by a photodiode device that transforms the data into forms that are readable and readily-consumed by end users

Li-Fi stands for Light Fidelity which is very much similar to conventional Wi-Fi, but uses visible light for data transfer instead of radio waves.

How is it related to Quantum mechanics?

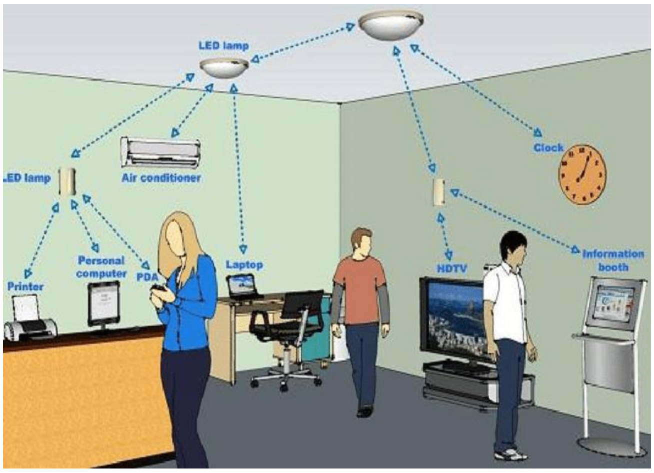
1. **Photon Behavior:**

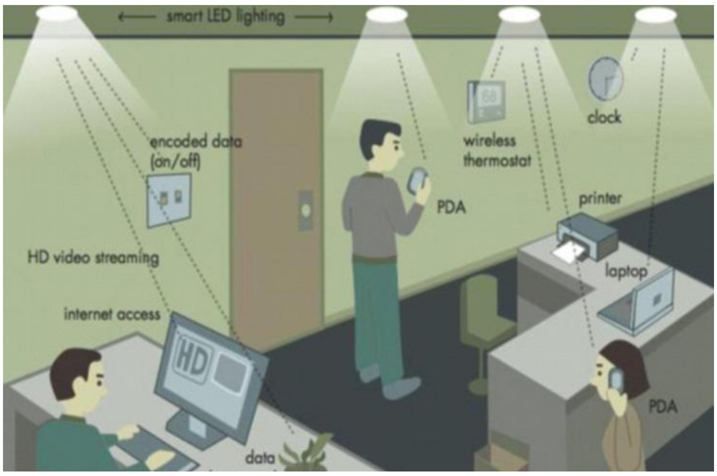
VLC uses light (photons) to transmit data. Photons fall in the domain of Quantum Mechanics, which describes the photons' wave-particle duality and energy quantization, essential for light modulation and detection in Li-Fi.

2. **Material Efficiency:**

VLC relies on LEDs and photodetectors. Quantum Mechanics governs the electronic properties of semiconductor materials, optimizing them for efficient light emission and detection.

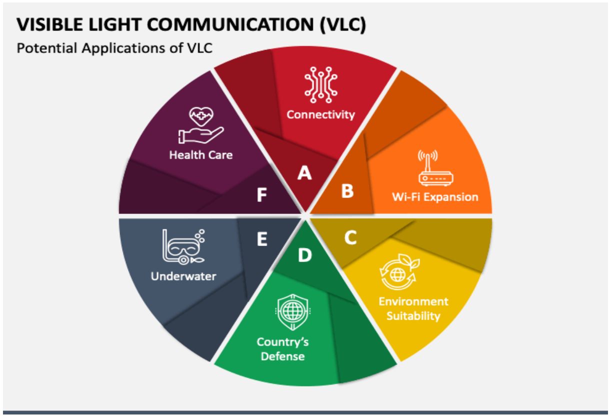
In essence, quantum mechanics provides the fundamental principles that enable the precise control, efficiency, and potential future advancements of VLC technology.





Problem Statement

- Our project aims to provide an environment free of radio waves, which on prolonged exposure are harmful, and an alternative method for data transfer where radio waves are not allowed.
- To utilize already existing infrastructure of LED lights and doubling their case of use, as a form of data transfer as well as source of light.



Major takeaways from current research

- **Data Rate Enhancements: Various research efforts have aimed at optimizing data rates. HAAS et al. (2016) reported achieving over 10 Gbps in laboratory settings.**
- **Signal Processing and Algorithms: Advanced signal processing techniques and algorithms have been developed to enhance Li-Fi performance, as documented by Pathak et al. (2015).**
- **Hybrid Systems: Studies on the coexistence of Li-Fi and Wi-Fi, such as those by Ayyash et al. (2016), have explored hybrid systems to combine the strengths of both technologies for enhanced communication networks.**
- **Modulation and Coding: Significant advancements have been made in modulation and coding techniques to enhance Li-Fi performance. Studies by Haas et al. (2011) explored the use of OFDM and Discrete Multi-Tone (DMT) modulation to achieve high data rates.**
- **Prototyping and Experimentation: Various prototypes and experimental setups have demonstrated Li-Fi feasibility. For instance, Komine and Nakagawa (2004) investigated smart lighting integration**

Objective

Our objective was to create a prototype which demonstrates the transfer of data utilizing light as the medium of transfer (data in the form a text or an image).
And to empower the importance of visual light data transfer in wireless communication to our community .

METHODOLOGY

This project focused on transmitting data using a laser and a laser sensor module. The development process was divided into four main steps to ensure efficient data transmission and reception.

Step 1: Image to Base64 Conversion

The first step involved converting a given image into a base64 encoded string. This was done using a Python script. The script read the image file in binary mode and encoded the binary data into a base64 string using the `base64` library. This base64 string was saved for subsequent transmission.

Step 2: Transmitting Base64 Data via Laser

Next, the Arduino was set up to transmit the base64 encoded string using a laser. The hardware setup included an Arduino board connected to a laser. The Arduino was programmed to send the base64 string bit by bit through the laser. Each character of the string was transmitted as binary data, represented by the laser being turned on or off for each bit.

Step 3: Receiving Data via Laser Sensor

The third step involved receiving the transmitted base64 data using a laser sensor module. An Arduino board was connected to the laser sensor. The Arduino was programmed to detect the laser signals and reconstruct the base64 string from these signals. The received base64 string was then printed to the serial monitor for verification.

Step 4: Converting Base64 to Original Format

The final step was converting the received base64 string back to its original format. A Python script was used to decode the base64 string back into its original binary form. This binary data was saved as an image file or text file using file handling operations in Python. The integrity of the received data was verified by comparing the original and decoded files, ensuring that the transmission was successful and accurate.

Prototype 1

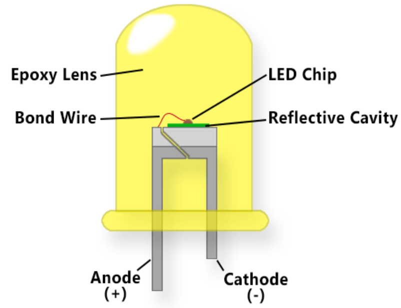
LED Module

Components used:

1. Arduino UNO micro-controller:

Arduino UNO (microcontroller board), with 14 digital I/O pins. Here we have used Arduino for converting the input data into a binary code and using this to pulse width modulate led/laser. We also used it to decode the received binary data.

2. LED (Light emitting diode): It is a P-N junction semiconductor diode that emits light when forward biased with an electric power supply.This is used as a transmitter to flicker according to the binary code being sent.A 220 ohm resisitor is used along with it to prevent damage to the LED due to any over current

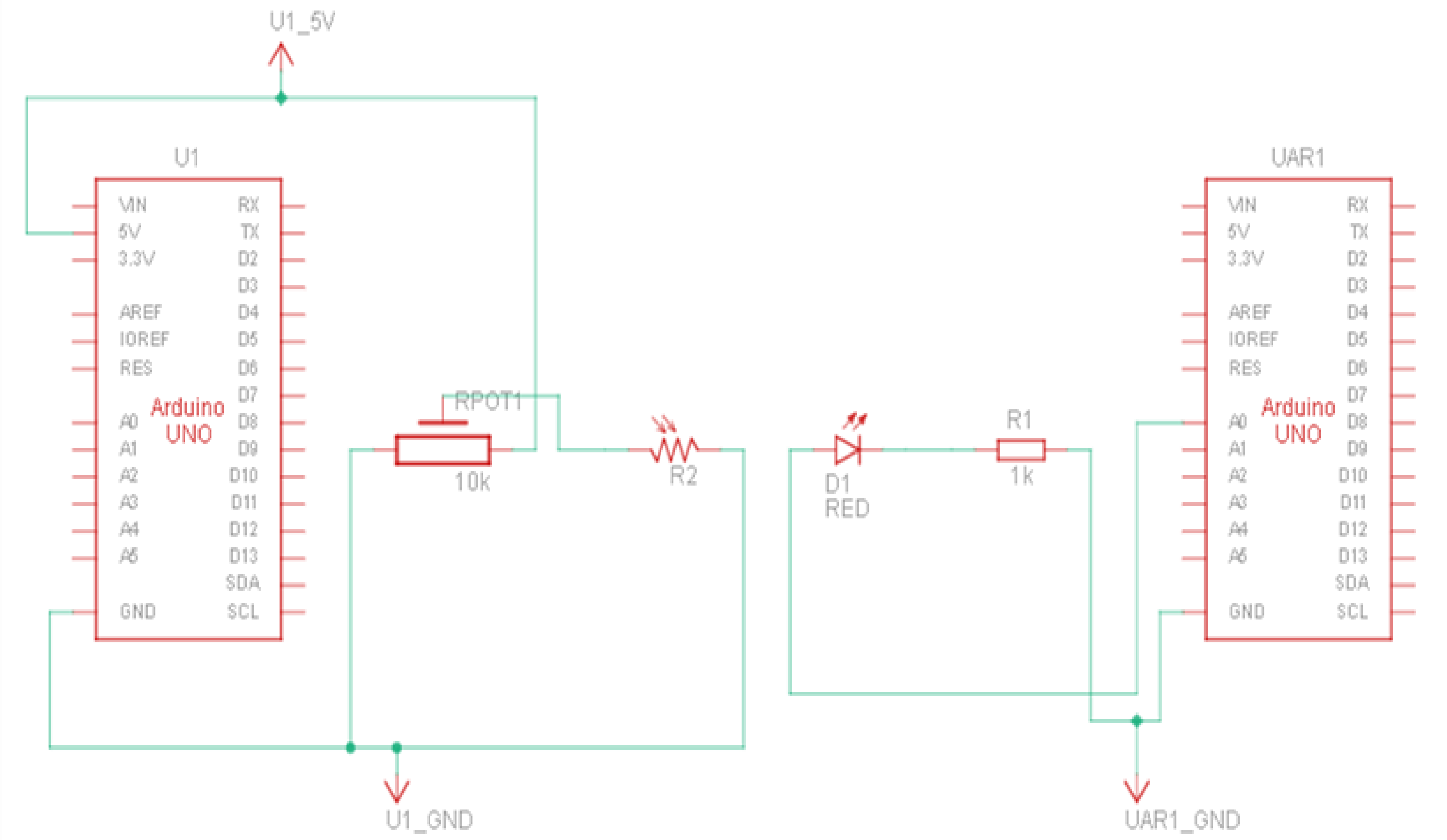


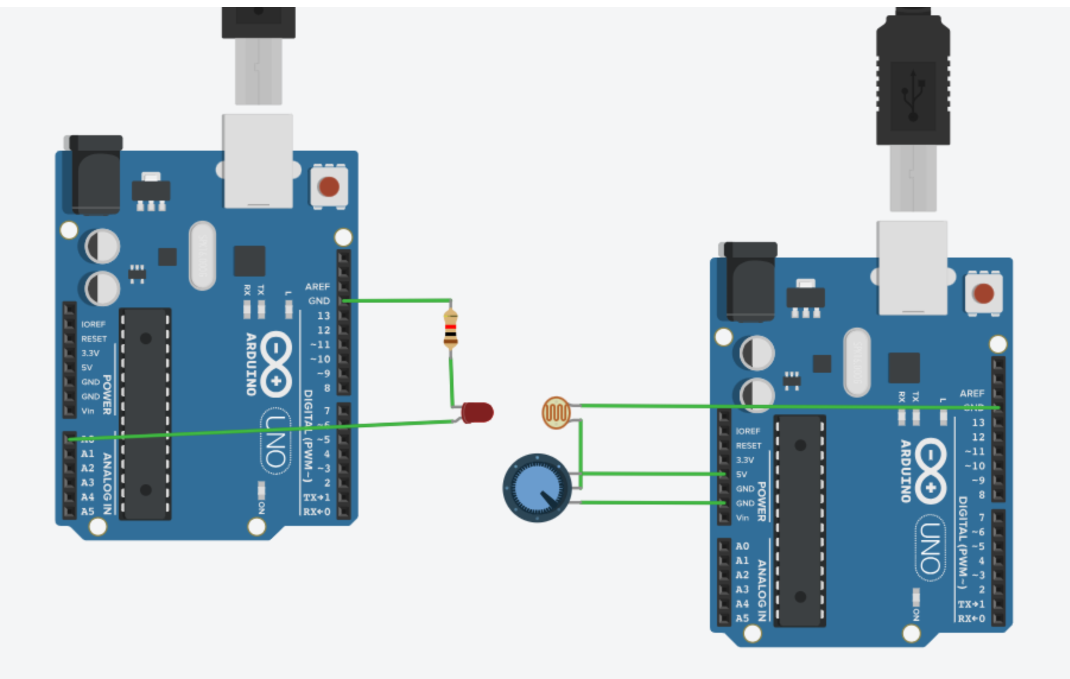
3. LDR (Light dependent resistor): It is a light sensitive resistor whose electrical resistance varies depending on the intensity of light it receives.it has negative resistance coefficient i.e., the brighter the light the lower the resistance.It gives an output of 0-5V if any light is detected in the form of binary ascii value. The initial reading(no light) of LDR is used as a threshold, when light is introduced this value drops detecting a light source.

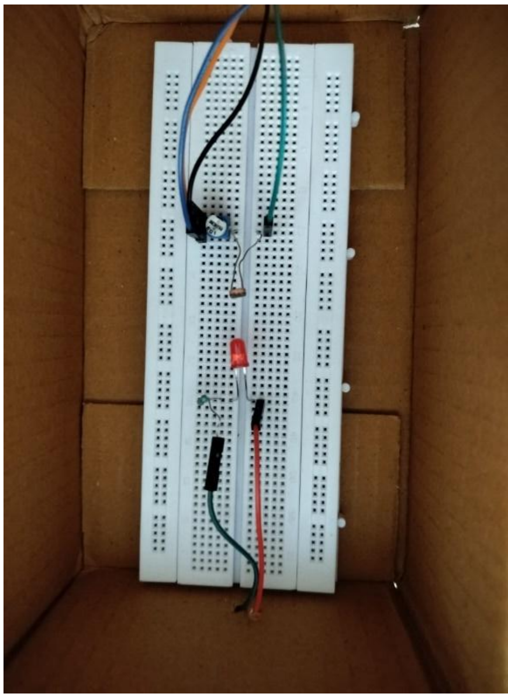
4. Potentiometer: It is a resistor device whose resistance can be varied according to the users need. The potentiometer we have used us of 10 kΩ resistance. This is used to control the output value of LDR, so as to identify if any light emitted from led.

Prototype 2

Circuit diagram:







Our model

Arduino CODE (transmitter):

```
#define led A0
#define period 100

char* text="//RVCE//";
int len=strlen(text);
void setup()
{
  // put your setup code here, to run once:
  pinMode(led,OUTPUT);
  Serial.begin(250000);
}

void loop()
{
  // put your main code here, to run repeatedly:
  for(int i=0;i<len;i++)
  {
    send_byte(text[i]);
  }

  delay(1000);

}

void send_byte(char my_byte)
{
  digitalWrite(led,LOW);
  delay(period);

  for(int i=0;i<8;i++)
  {
    digitalWrite(led, (my_byte & (0x01<<i))!=0);
    Serial.println(my_byte & (0x01<<i));
    delay(period);
  }

  digitalWrite(led,HIGH);
  delay(period);

}
```

In the above code the LED is given analog input for the purpose of pulse width modulation to send the binary ascii value of each character in the text “//RVCE//”. The text is sent repeatedly with a gap of 1 second between each loop. And each character in the text sent at a speed of 100 millisecond.

Receiver CODE

```
#include <stdio.h>
#define ldr A0
#define threshold 50
#define period 1

bool previous_state;
bool current_state;

void setup()
{
  // put your setup code here, to run once:
  Serial.begin(9600);
}

void loop()
{
  current_state=get_ldr();
  if(!current_state && previous_state)
  {
    print_byte(get_byte());
  }
  previous_state=current_state;

}
bool get_ldr()
{
  //to check if the laser is on or off
  int voltage=analogRead(ldr);
  return voltage>threshold ? true:false;
}

char get_byte()
{
  //to extract binary code sent by the laser
  char ret=0;
  delay(period*1.5);
  for(int i=0;i<8;i++)
  {
    ret=ret|get_ldr() << i;
    delay(period);
  }
  return ret;
}

void print_byte(char my_byte)
{
  //extracts the character specific to the ascii binary code received and prints it
  char buff[2];
  sprintf(buff,"%c",my_byte);
  Serial.print(buff);
}
```

The receiver part uses an LDR to receive the binary data sent using PWM, and decodes each ascii value to its character and prints simultaneously.

Although the LED module worked, it wasn't able to transfer text at the speed we expected. Hence, we upgraded to the laser module.

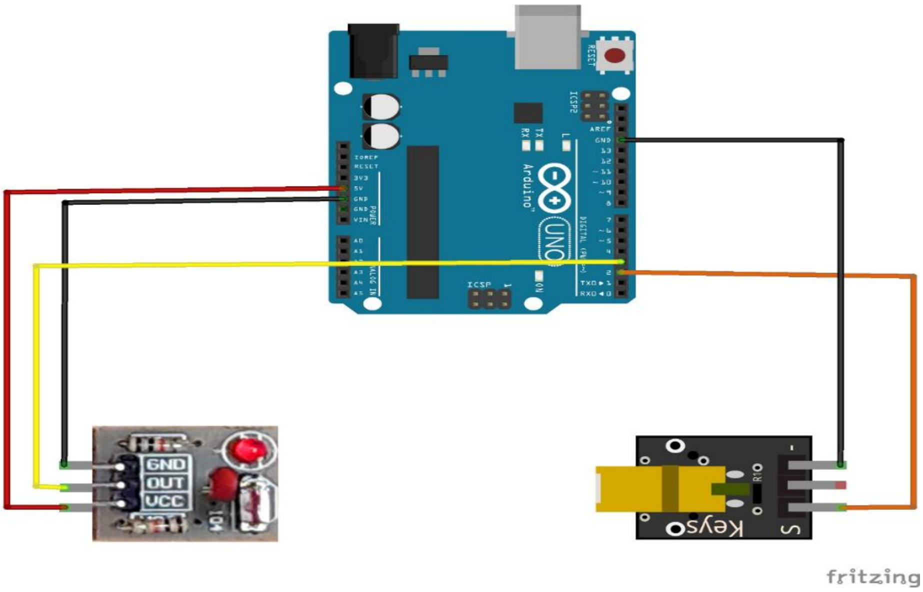
LASER Module

Components used:

1. KEYES(KY-008) LASER module:

The KY-008 receiver is highly sensitive to light of wavelength 650-680nm (i.e., only to redlight) and can perceive sudden change in wavelength which makes it ideal for faster data transfer for a longer range than LED.
The transmitter is also highly sensitive and can emit light with sharp intensity changes, and flicker at a faster rates of about 1 millisecond

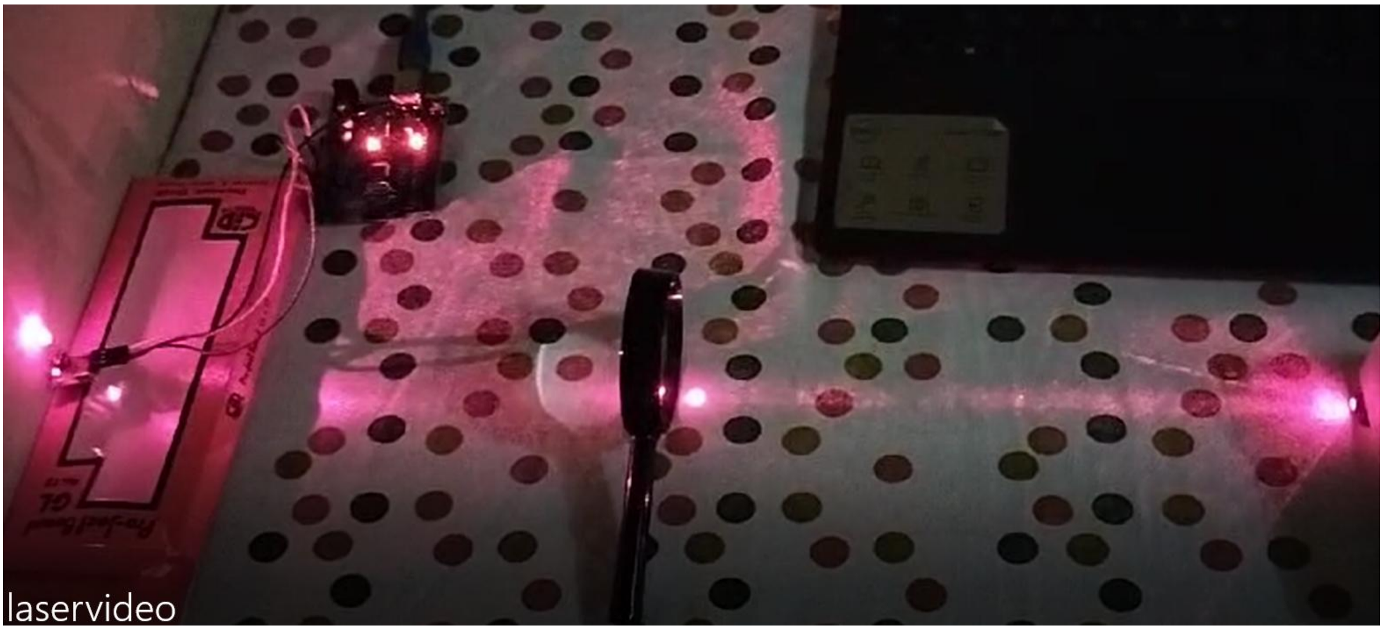
Circuit diagram:



Transmitter code

```
#define led 6
#define period 1

char* text="Hey, this is a Visual light communication(VLC) project!";
int len=strlen(text);
void setup()
{
  // put your setup code here, to run once:
  pinMode(led,OUTPUT);
  Serial.begin(250000);
}

void loop()
{
  // put your main code here, to run repeatedly:
  for(int i=0;i<len;i++)
  {
    send_byte(text[i]);
  }

  delay(5000);

}
void send_byte(char my_byte)
{
  //transmittes the binary code for each character in the text using PWM
  digitalWrite(led,LOW);
  delay(period);

  for(int i=0;i<8;i++)
  {
    //LED emmits light only when the bit reads one
    digitalWrite(led, (my_byte & (0x01<<i))!=0);
    Serial.println(my_byte & (0x01<<i));
    delay(period);
  }

  digitalWrite(led,HIGH);
  delay(period);

}
```

The receiver code for the laser module is same as that of the LED module, but with a period of 1 milli second, i.e., each character of the text is sent and received at a speed of 1 millisecond.



Now we had achieved high speed text transfer so we further extended our thoughts for image transfer.

For this we followed an algorithm:

- First encode the image to its "base64" using a python library called base64. A base 64 file of an image is nothing but a text file of an image.
- This base64 file is sent as a text using the LASER module from the above code
- This base64 file received is then decoded back to its image and stored in a folder using python library.

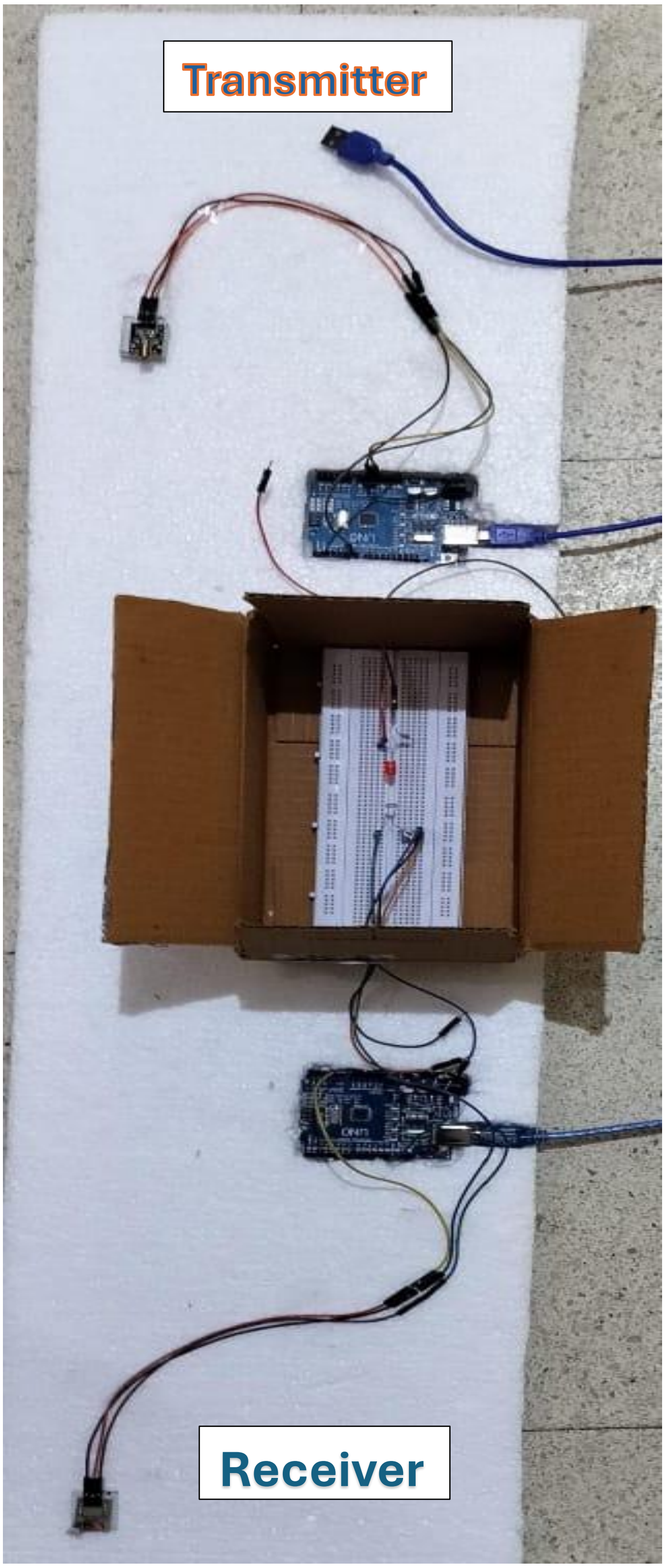
Python code to encode an image file to "base64" file

```
import base64
def image_to_base64(image):
    with open(image, 'rb') as image_file:
        base64_string=base64.b64encode(image_file.read()).decode('utf-8')
        return base64_string
image="C:\\Users\\THEJAS\\OneDrive\\Desktop\\EL\\images.png"
base64_string=image_to_base64(image)
print(base64_string)
```

Python code to decode the received base64 file to its image

```
import serial
import time
import base64
import io
from PIL import Image
import binascii

fixed_base64_str="put the Base64 text of image here"
image_bytes=base64.b64decode(fixed_base64_str)
with open("extra.png","wb") as f:
    f.write(image_bytes)
```

FINAL PROTOTYPE



RESULTS and DISCUSSION

- We were able to flicker with an experimental frequency of 1000Hz and transfer a low-resolution picture (360 x 360 pixels) in a matter of 5 seconds
- We were able to transfer any text within 2 seconds with the LASER module
- Only disadvantage of wireless VLC is that the transmitter and receiver need to be in-line with each other
- If we were to implement VLC in real-life we would require an LED that flickers at a faster rate than the human eye can perceive and a receiver sensitive enough to perceive such fast incoming data

Challenges and Insights:

Our Li-Fi prototype successfully demonstrated the feasibility of transmitting images and text data using light. However, the project also encountered a key challenge: speed limitations. Due to the constraints of the available electronics, the data transfer rate was restricted to approximately 1 millisecond (ms) per bit. This limitation translates to slower overall transmission times for images and text data. This experience underscores the crucial importance of carefully considering both the capabilities of the light source and the supporting electronics when designing Li-Fi systems for optimal performance.

Enhancing Transmission Speed: A Strategic Approach

Despite this initial hurdle, the project has opened exciting avenues for further exploration. Here are some potential strategies to overcome the speed limitation and achieve faster Li-Fi transmission speeds:

- **Hardware Optimization:** Upgrading the core electronic components could significantly improve data transfer rates. Utilizing higher-performance microcontrollers equipped with faster processors would facilitate more efficient data transmission and reception by the Arduino Uno or a similar microcontroller. Additionally, employing dedicated laser driver modules designed specifically for high-speed data transmission could provide more precise control over the laser's behaviour, optimizing light pulse generation for faster communication.

- **Advanced Laser Sensor Selection:** Research into alternative laser sensor technologies can potentially unlock transmission speeds in the microsecond (μs) or even nanosecond (ns) range. Here are some promising options:
    - **Avalanche Photodiodes (APDs):** APDs offer increased sensitivity compared to standard photodiodes. This enhanced sensitivity allows for the detection of weaker light signals, potentially enabling faster data transmission with lower power lasers, leading to improved overall efficiency.
    - **Photomultiplier Tubes (PMTs):** PMTs represent a technology capable of even higher sensitivity than APDs, making them potentially suitable for achieving ultra-fast data transmission in scenarios with minimal light levels. However, PMTs are typically more expensive and require specialized operating conditions, making them a less immediately viable option.
- **Advanced Modulation Techniques:** While hardware upgrades are pursued, exploring advanced modulation techniques like Manchester encoding can optimize data transmission within the current hardware limitations. By incorporating additional information into the signal itself, Manchester encoding has the potential to improve data integrity and potentially increase transmission efficiency even with the 1ms constraint.

By addressing these challenges and investigating cutting-edge technologies, we can unlock the full potential of Li-Fi for high-speed data transmission. This, in turn, will pave the way for the development of innovative applications in a wide range of fields

Conclusion

In a world where data is everything and the advances in radio wave technology is reaching its limit, VLC provides a very cost effective, Radio wave free and most important faster method of transferring the data.

We believe that Wi-Fi and Li-Fi can co-exist, and such a hybrid system is the future of data transfer and communication industry.

THANK YOU